Group: 2

Internal Pipe Climber (IPC) - Checklist

#	Rules and regulation	How it was checked	Check
1	Size of the model	*Length：25.3cm width: 8.5-16.1 height: 8.7cm*	√
2	Type of batteries	*According to the battery requirements of the competition18650 batteries.*	√
3	The starting and Final positions of the device will be determined from the datum point (datum to ground)	*The height given by the pipeline Height:1m*	√
4	Any repairs must be completed within the given timeframe, either before or during the specified period.	*Successfully complete the tasks of design, modeling, parts procurement, assembly, and testing in a sequential manner. Each trial is conducted with precision and achieves the desired outcomes.*	√
5	Advanced Challenge – Programmable devices are allowed; team must show evidence of independent coding	*The programming software utilized in our project includes C51 and C251, which are primarily employed for the identification of black lines. Relevant code examples can be found on the poster.*	√
6	The code does not include any measurements that require editing	*We employ infrared sensors to detect a black line, upon which the system will cease operation. This mechanism is not constrained by vehicle speed and does not require distance programming.*	√
7	No reprogramming is allowed between runs	*No manual intervention is needed throughout the entire operation of the vehicle.*	√
9	The device must not be operated by a remote control	*No remote control modules were added. The device was activated via a physical power switch and operated fully autonomously without human intervention during all test runs (video evidence provided).*	√
13	All safety-related items must always be on the device and must not come loose at any time during the competition.	*Inspected the fastening status of screws, adhesive tapes, and buckles. Conducted vibration tests (10 second of high-frequency shaking) to ensure no components loosened.*	√
14	The total cost of all components and the device itself must be RMB1000 or less.	*835.06*	√
15	The device must approach the length (1m) tube vertically While carrying the 0.1kg load.	*The vehicle was manually secured in a verified vertical pipe. A 500g iron chain was added during load testing, and stability was confirmed through repeated climbing experiments.*	√

16	The device must stop at the target line (forward)for between 3-5 seconds before heading off to the top line	*Infrared sensors were used to detect the forward target line. Upon triggering, a timer recorded a 4-second dwell time (within the 3-5s requirement). The delay function in the code was set to 4000ms and verified via serial output logs.*	√
17	The device must stop at the target line (backwards) for between 3-5 seconds before heading back to the start line	*Infrared sensors were used to detect the backward target line. Upon triggering, a timer recorded a 3.5-second dwell time (within the 3-5s requirement). The delay function in the code was set to 3500ms and verified via serial output logs.*	√